BubblesO2 All-In-One (AIO) System

IT Operations, Infrastructure Governance & Core ITSM Manual

Project Title: BubblesO2 All-In-One (AIO) System — IT Operations & Core Infrastructure Manual

User: Iskandar

Department: IT Department

Date: September 2026

Target Audience: Internal IT Support, System Administrators & Executive Management

for BubblesO2 SDN BHD

Prepared by,

Muhammad Iskandar Zulkarnain Bin Zulkifli
System Developer,
Information Technology,
Information Technology Department

Approved by,

Abdul Aziz Arifin
Manager,
Information Technology,
BubblesO2 Sdn Bhd

Copyrights reserved. The documentation is prepared solely for the use of management only. No part of it may be circulated or reproduced for distribution outside of the organization without prior consent of IT Department, BubblesO2. Prepared By Muhammad Iskandar

1.0 EXECUTIVE SUMMARY & ARCHITECTURAL OVERVIEW

The **BubblesO2 All-In-One (AIO) System** is an internally engineered, enterprise-grade cloud application built to unify daily facility operations, digital production reporting, and comprehensive **Information Technology Service Management (ITSM)** under a single, highly performant web interface.

Rather than competing with the company's tier-1 financial ERP (Sage X3 V12), the AIO platform serves as the **agile operational front-end**. It captures ground-level shopfloor transactions, hardware maintenance tickets, asset audits, and shift handovers in real time, eliminating paper clutter and manual handwriting calculations while ensuring that all upstream data is verified, authenticated, and auditable.

1.1 Core Platform Technology Stack

ARCHITECTURE LAYER	TECHNOLOGY COMPONENT	OPERATIONAL FUNCTION & RELIABILITY GUARANTEE
Web Application Tier	Next.js 16 (App Router) & React 19	Server-side rendering (SSR) for instantaneous page loads, zero client lag, and seamless mobile responsiveness on factory tablets.
Database & ORM Tier	PostgreSQL with Prisma ORM 7	High-performance relational persistence with connection pooling (`@prisma/adapter-pg`), transaction isolation, and foreign key integrity.
Security & Access Control	NextAuth.js v5 with RBAC & bcryptjs	Cryptographically signed session cookies, salt-hashed passwords, and strict two-tier role & module permission matrices.
Document & Object Storage	Cloudflare R2 / AWS S3 API	S3-compatible immutable cloud storage for digital employee signatures, PO uploads, and PDF reports.
Integration Gateways	Google Calendar API v3 & Resend API	Two-way synchronization of maintenance schedules and sub-second dispatch of transactional ticket alert emails.

2.0 IT OPERATIONS DASHBOARD & ANALYTICS (/DASHBOARD)

The IT Operations Dashboard functions as the centralized mission control for the Information Technology Department, presenting immediate, real-time telemetry across ticketing throughput, hardware deployments, and operational schedules.

2.1 Executive Metric Telemetry Cards (Real-Time Overview)

TOTAL TICKETS	PENDING ISSUES	RESOLVED TODAY	ACTIVE ASSETS	UPCOMING EVENTS
All-Time	**Active**	**Daily Turn**	**Hardware**	**Next 7 Days**
Cumulative Log	Needs Attention	SLA Compliant	Live in Service	Preventive Window

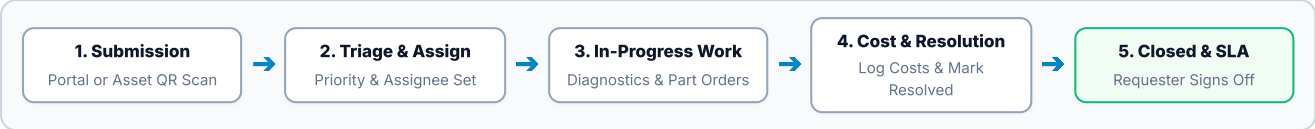
2.2 Dynamic Analytics Telemetry

- **Status Distribution (Donut Breakdown):** Visualizes current backlog distribution across Open, In-Progress, Resolved, and Closed states.
- **Departmental Incident Concentration:** Identifies which branch or operating unit generates the highest ticket volume, isolating chronic hardware failures.
- **Recent Activity Audit Trail:** Logs the latest five cross-system actions (e.g. Asset checkouts, user status modifications, maintenance completions).

Copyrights reserved. The documentation is prepared solely for the use of management only. No part of it may be circulated or reproduced for distribution outside of the organization without prior consent of IT Department, BubblesO2. Prepared By Muhammad Iskandar

3.0 IT SMART TICKETING SYSTEM (ITSM LIFECYCLE)

The IT Smart Ticketing System provides a standardized, auditable framework for employees across all branches (Rantau Factory, Sendayan, and HQ) to report technical disruptions, hardware faults, and network failures.



3.1 Step-by-Step Ticket Creation & Input Field Specifications:

FIELD NAME	INPUT TYPE	REQUIREMENT	OPERATIONAL GUIDELINES & SYSTEM HANDLING
Requester Name	Text String	**Mandatory**	Full legal name of the employee logging the issue. Pre-filled automatically if user is authenticated.
Contact Phone	Phone Number	**Mandatory**	Direct contact number or WhatsApp extension for rapid on-site technician dispatch.
Branch & Unit	Cascading Dropdowns	**Mandatory**	Select the affected physical facility (e.g. *BubblesO2 Rantau*) and unit (e.g. *Supervisor Production*).
Asset Tag Link	Searchable Selector	Optional	Search and select the affected asset tag (e.g. `TH47RF600H`). Automatically links hardware specifications.
Issue Title	Text String	**Mandatory**	Concise, descriptive headline of the failure (e.g. *"HP OfficeJet 9130 Printer Cannot Print DO"*).
Detailed Description	Textarea	**Mandatory**	Thorough explanation of the failure symptoms, error codes, and preliminary troubleshooting performed.
Priority Classification	Enum Selection	**Mandatory**	**LOW:** General inquiry | **MEDIUM:** Single user impacted | **HIGH:** Department halted | **CRITICAL:** Plant line down.

3.2 Technical Resolution, Cost Attribution & Mean Time to Repair (MTTR)

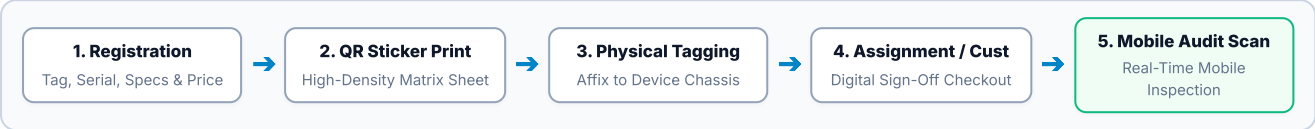
Upon taking ownership of an active ticket, the assigned IT engineer executes the following protocol:

- **Transition to In-Progress:** Automatically alerts the user that a technician is actively investigating.
- **Hardware Spare Parts Costing:** If replacement parts are consumed (e.g. thermal printhead, SSD, patch cord), the engineer selects the part from the **Cost Catalog**. The system tallies material costs against the requester's departmental budget.
- **Resolution Sign-Off:** Documenting the technical root cause and applied fix (e.g. *"Replaced damaged USB print interface and updated firmware"*). Marking as `RESOLVED` timestamps completion for SLA auditing.

Copyrights reserved. The documentation is prepared solely for the use of management only. No part of it may be circulated or reproduced for distribution outside of the organization without prior consent of IT Department, BubblesO2. Prepared By Muhammad Iskandar

4.0 ASSET MANAGEMENT & QR CODE LIFECYCLE

The Asset Management module delivers comprehensive cradle-to-grave tracking of corporate physical assets, encompassing ICT hardware, production equipment, electrical assets, safety tools, and office machinery.



4.1 Asset Registration Form Field Specifications (/dashboard/assets/new):

FIELD NAME	DATA TYPE	REQUIREMENT	TECHNICAL STANDARD & FORMATTING RULES
Asset Tag	Alphanumeric	**Mandatory & Unique**	Standardized corporate asset tag (e.g. `ICT-LPT-0045`, `PRN-HP-0012`). Enforced unique index at DB level.
Asset Name	Text String	**Mandatory**	Full descriptive name (e.g. *"Lenovo ThinkPad E14 Gen 4 Core i7"*).
Asset Category	Enum Selection	**Mandatory**	`ICT_EQUIPMENT`, `OFFICE_EQUIPMENT`, `MACHINE`, `ELECTRICAL_EQUIPMENT`, `TOOLS`, `SAFETY`.
Brand & Model	Text String	Optional	Hardware OEM and model designation (e.g. *Lenovo / 21E3006CMY*).
Serial Number	Text String	**Recommended**	Chassis OEM serial number. Crucial for vendor warranty claims and insurance audits.
Device ID / Hostname	Text String	Optional	Network hostname or static IP address (e.g. `BO2-IT-NB04`).
Purchase Date & Cost	Date / Decimal	Optional	Acquisition date and purchase price in MYR for depreciation and capital tracking.
Department & Location	Foreign Keys	**Mandatory**	Physical location (Block, Room, State) and organizational department owner.

4.2 QR Code Printing & On-Site Mobile Scanning Engine

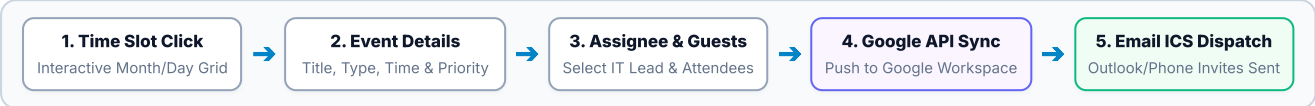
> **Sticker Generation Client (/dashboard/assets/print-qr):**
> The system dynamically generates scalable vector 2D QR codes encoded with the asset's public verification URL (`/public/asset/{id}`). The print layout is optimized for standard Avery / Brother thermal synthetic waterproof sticker sheets.

Mobile Camera Scanner: Field technicians open `/dashboard/assets/audit` on any smartphone or tablet. Scanning the physical tag immediately displays live equipment status, assigned user, maintenance history, and allows 1-tap logging of maintenance requests.

Copyrights reserved. The documentation is prepared solely for the use of management only. No part of it may be circulated or reproduced for distribution outside of the organization without prior consent of IT Department, BubblesO2. Prepared By Muhammad Iskandar

5.0 EVENT CALENDAR & OPERATIONAL SCHEDULER (/CALENDAR)

The Event Calendar coordinates scheduled maintenance windows, network downtime periods, vendor demonstrations, and internal IT team deployment schedules, preventing operational conflicts across departments.



5.1 Event Scheduling Modal Field Specifications:

FIELD NAME	DATA TYPE	REQUIREMENT	TECHNICAL HANDLING & DESCRIPTION
Event Title	Text String	**Mandatory**	Brief headline of the operational event (e.g. *"Server Infrastructure Firmware Patching"*).
Start & End Time	DateTime	**Mandatory**	Precision timestamp. Evaluated in Malaysia Time (`Asia/Kuala_Lumpur`, UTC+8).
Category / Tag	Enum Selector	**Mandatory**	`MAINTENANCE`, `MEETING`, `PROJECT`, `TRAINING`, `HOLIDAY`, `CRITICAL`.
Primary Assignee	User Selector	**Mandatory**	Designated IT Engineer responsible for executing and closing the scheduled task.
Attendee Roster	Multi-Select	Optional	Cross-departmental staff invited. System auto-generates calendar invite attachment (`.ics`).
Google Calendar Sync	Boolean Toggle	Optional	When toggled, system executes service account OAuth handshake with Google Calendar API.

5.2 Two-Way Google Workspace Calendar API Synchronization

Automated Cloud Handshake:
The application connects securely to Google Workspace via a dedicated Google Service Account (`GOOGLE_CLIENT_EMAIL` and private RSA key). When an event is scheduled or modified, the system creates or updates the event in the corporate primary calendar (`GOOGLE_CALENDAR_ID`), providing real-time calendar synchronization to mobile devices.

Copyrights reserved. The documentation is prepared solely for the use of management only. No part of it may be circulated or reproduced for distribution outside of the organization without prior consent of IT Department, BubblesO2. Prepared By Muhammad Iskandar

6.0 USER MANAGEMENT & ROLE-BASED ACCESS CONTROL (RBAC)

User access governance in the BubblesO2 AIO platform is secured through a robust two-layer security framework: **Hierarchical System Roles** and **Granular Functional Module Flags**.

6.1 Hierarchical Role Permission Matrix (/users)

ROLE CODE	ROLE TITLE	PERMITTED SYSTEM PRIVILEGES	OPERATIONAL TARGET AUDIENCE
SUPERADMIN	Master System Admin	Full read/write rights across all branches, user creation, role assignment, system settings, and purge tools.	IT System Developers, IT Manager.
ADMIN	Department Administrator	Full operational management over assigned department, asset assignment, and ticket resolution.	Assistant Managers, Department Supervisors.
DIRECTOR	Executive Management	Company-wide read-only access to analytics dashboards, production KPIs, and audit logs.	Managing Director, Board Members.
HOD	Head of Department	Approval authority over department shift reports, asset requisitions, and maintenance expenses.	Head of Production, Head of Warehouse.
LEADER	Line / Shift Leader	Authority to verify daily shift production forms, perform shift handovers, and apply digital approval signatures.	Plant Shift Supervisors, QA Team Leads.
STAFF	Standard Operational User	Submitting daily production data, logging IT helpdesk tickets, viewing personal work history.	Machine Operators, Forklift Drivers, Clerks.
DRIVER	Logistics & Dispatch	Focused view for Distribution Orders (DO), dispatch check-ins, and returnable pallet tracking.	Delivery Drivers, External Transporters.

6.2 Granular Functional Module Permission Array

In addition to their role, each user record maintains a `modules` array string. If a module tag is omitted, the corresponding navigation tab is dynamically removed from the user's interface:

- `TICKETS`: Grants access to submit, track, and monitor IT service tickets.
- `ASSETS`: Grants access to hardware registry, QR sticker generation, and physical audits.
- `FORMS`: Grants access to daily production logs (KAR, DPR) and SGMP maintenance forms.
- `WAREHOUSE`: Grants access to DO receiving, Warehouse Gate Pass, and returnable pallet tracking.
- `SALES`: Grants access to Customer master directories, AI document intake, and commercial Sales Orders.

6.3 Digital E-Signature Capture & Compliance Protocol

> **Paperless Regulatory Sign-Off:**
> In compliance with food manufacturing standards (HACCP, GMP), supervisors must apply digital signatures to lock production reports. In `/users`, clicking the **[Signature]** modal opens a touch-sensitive canvas. The captured vector is converted to PNG, uploaded to Cloudflare R2 storage, and mapped to the user record for instant automated application to approved documents.

Copyrights reserved. The documentation is prepared solely for the use of management only. No part of it may be circulated or reproduced for distribution outside of the organization without prior consent of IT Department, BubblesO2. Prepared By Muhammad Iskandar

7.0 ORGANIZATIONAL HIERARCHY & COST CATALOG (/ORGANIZATION)

The Organization module models the multi-site enterprise hierarchy of BubblesO2 Sdn Bhd, providing clean data segregation across branches, operating units, and physical facilities.



7.1 Managing Branches & Operational Units

ENTITY LEVEL	REQUIRED FIELDS	OPERATIONAL RULES & CONSTRAINTS
Branch (Facility)	Branch Name (String), Location Address (Text)	Represents separate physical compounds. Deleting a branch is rejected by database constraints if child units exist.
Operating Unit	Unit Name (String), Parent Branch (Foreign Key)	Represents functional work centers within a facility. Every user and ticket is mapped to an operational unit.

7.2 Hardware Replacement Cost Catalog (/organization)

To ensure transparent departmental accountability, IT maintains a standardized catalog of replacement hardware and servicing fees:

CATALOG ITEM NAME	STANDARD UNIT PRICE (MYR)	TYPICAL APPLICATION SCENARIO
NVMe SSD 512GB High-Speed	RM 185.00	Replacing failed mechanical hard drives or workstation storage upgrades.
DDR4 16GB Workstation RAM	RM 140.00	Performance upgrades for factory production monitoring terminals.
Thermal Print Head 4-Inch	RM 320.00	Warehouse barcode printer replacement for DO and pallet labeling.
CAT6 Shielded Patch Cord 3M	RM 25.00	Resolving physical network drops at packaging lines.
Network Switch 8-Port Gigabit	RM 160.00	Expanding terminal connectivity in warehouse dispatch offices.

Copyrights reserved. The documentation is prepared solely for the use of management only. No part of it may be circulated or reproduced for distribution outside of the organization without prior consent of IT Department, BubblesO2. Prepared By Muhammad Iskandar

8.0 SYSTEM SETTINGS & GLOBAL CONFIGURATIONS (/SETTINGS)

The Settings module centralizes enterprise system controls, automated maintenance schedulers, communication gateways, and corporate broadcast notices.

8.1 Corporate Broadcast Notice Board

Superadmins can broadcast critical operational notices that render at the top of every employee's dashboard upon login:

- **Maintenance Downtime Notices:** e.g. *"Server Infrastructure Upgrade: Saturday 10:00 PM – 12:00 AM. System offline."*
- **SOP Updates:** e.g. *"Revision 2 of Warehouse Gate Pass Policy now active. Please review."*
- **Priority Flagging:** Urgent notices render with an amber alert container to ensure mandatory employee acknowledgement.

8.2 Dual Email Gateway Architecture

GATEWAY SERVICE	CONFIGURATION KEYS	WORKLOAD DISTRIBUTION
Resend API	`RESEND_API_KEY`	Dispatches transactional ticket alerts, password resets, and automated daily shift digest summaries.
Corporate SMTP	`SMTP_EMAIL`, `SMTP_PASSWORD`	Dispatches calendar meeting invites with native `.ics` MIME attachments to Outlook and Google Calendar.

9.0 IT SUPPORT FAQ & TROUBLESHOOTING MATRIX

REPORTED INCIDENT / ERROR	PROBABLE ROOT CAUSE	IMMEDIATE IT RESOLUTION PROCEDURE
"Access Denied: Module Not Authorized"	User profile is missing the required module flag in their database record.	Open `/users` as Superadmin, edit the user, enable the target module checkbox (e.g. `ASSETS`, `TICKETS`), and save.
"Camera will not activate in Asset QR Scanner"	Web browser camera permissions blocked, or session is running over insecure HTTP.	Ensure connection is HTTPS (`https://...`). In Chrome/Safari mobile settings, grant camera permissions.
"Ticket status will not change to Closed"	User has not accepted resolution, or required cost logging is pending.	Ensure an IT technician has logged resolution notes and clicked Resolved before final sign-off.
"Calendar event not syncing with Google Calendar"	Google private key invalid or corporate calendar ID incorrect.	Verify `GOOGLE_PRIVATE_KEY` and `GOOGLE_CLIENT_EMAIL` in deployment environment variables.
"Duplicate Asset Tag error on registration"	Asset Tag identifier is already registered to an active or retired device.	Verify existing asset tag in `/dashboard/assets/list`. If decommissioned, update existing record.

Copyrights reserved. The documentation is prepared solely for the use of management only. No part of it may be circulated or reproduced for distribution outside of the organization without prior consent of IT Department, BubblesO2. Prepared By Muhammad Iskandar

APPENDIX A: STEP-BY-STEP UI TUTORIAL WALKTHROUGH

Step 1 — Accessing the IT Smart Ticketing Portal

Users navigate to the top navigation bar and click **Tickets**. The dashboard displays all open, in-progress, and resolved service requests.

> **Action:** Click the blue **[+ New Ticket]** button in the top right corner (or mobile floating **[+]** action button) to launch the ticket submission modal.

Step 2 — Completing the Incident Requisition Form

> **Critical Fields to Complete:**
> - **Requester & Phone:** Enter direct contact details.
> - **Branch & Unit:** Specify facility location.
> - **Asset Tag Link:** If hardware is involved, select the asset tag to speed up diagnostics.
> - **Priority Selection:** Select appropriate urgency level.
> - Click **[Submit Ticket]** to dispatch immediate email notifications to the IT team.

Step 3 — Registering Physical Assets & Generating QR Stickers

To add a new device to the corporate inventory:

1. Navigate to **Assets → New Asset** (`/dashboard/assets/new`).
2. Fill in the standardized Asset Tag (e.g. `ICT-LPT-0045`), Serial Number, Brand, Category, and Assigned Location.
3. Click **[Create Asset]**.
4. Navigate to **Print QR Stickers** (`/dashboard/assets/print-qr`), select the asset, and print the waterproof sticker.
5. Affix the sticker securely to the device chassis for mobile audit scanning.

Step 4 — Scheduling Team Events & Maintenance on the Calendar

To coordinate operational tasks:

1. Navigate to **Calendar** (`/calendar`).
2. Click directly on the target date or hourly time block.
3. Enter the event title, category, start/end timestamps, and assign the lead technician.
4. Select attendees to receive automated email calendar invites (`.ics`).
5. Click **[Schedule Event]**. The task immediately appears across all team schedules.

Copyrights reserved. The documentation is prepared solely for the use of management only. No part of it may be circulated or reproduced for distribution outside of the organization without prior consent of IT Department, BubblesO2. Prepared By Muhammad Iskandar

10.0 CONCLUSION & FORMAL SIGN-OFF

The **BubblesO2 All-In-One (AIO) System** is more than a software utility; it is the operational nervous system connecting staff across our factory floors, logistics warehouses, and corporate offices. By establishing transparent IT ticketing, robust asset traceability, unified event scheduling, and strict access governance, the platform protects company investments, mitigates operational downtime, and enforces complete data integrity.

This technical manual serves as the definitive operating standard for the Information Technology Department. All internal IT engineers, support specialists, and system administrators are expected to adhere strictly to the protocols, field standards, and security procedures detailed herein.

DOCUMENT APPROVAL & FORMAL ISSUANCE:

PREPARED BY:

Muhammad Iskandar Zulkarnain Bin Zulkifli
System Developer, Information Technology
Information Technology Department
BubblesO2 Sdn Bhd
Date: September 2026

APPROVED BY:

Abdul Aziz Arifin
Manager, Information Technology
Information Technology Department
BubblesO2 Sdn Bhd
Date: September 2026

Copyrights reserved. The documentation is prepared solely for the use of management only. No part of it may be circulated or reproduced for distribution outside of the organization without prior consent of IT Department, BubblesO2. Prepared By Muhammad Iskandar